

Foundations of Arithmetic Power Geometry IX: Source Concentration Theory and Entropy-Controlled APG Density Bounds

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Abstract

This paper develops the ninth volume of Arithmetic Power Geometry (APG) by proving the source-concentration component of the APG spectral-conductor program. Earlier APG volumes introduced local entropy-controlled closure defects, information-geometric stabilization, integrated closure-defect functionals, scale-corrected invariants, discrete prime-chain regularization, conditional APG–Arakelov height coupling, discrepancy-source compatibility, and the Spectral–Green Bridge. The unresolved source-side obstruction was whether the deformation-induced density ρ_{APG} on the compactified modular curve $X = X_0(N)^*(\mathbb{C})$ could develop non-integrable mass concentration, especially near cusp neighborhoods. This paper gives a clean APG IX resolution by formulating the APG source as a Radon–Nikodym density on a cusp-regularized compact modular surface, proving a non-circular Hilbert-space decomposition

$$\rho_{\text{APG}} = \frac{1}{V} + \alpha_{\text{APG}}\sigma,$$

constructing a Hilbert–Schmidt transport kernel for the variance–defect transfer, proving the entropy-amplitude inequality

$$\alpha_{\text{APG}} \leq C_H H(W),$$

and deriving the cusp dampening factor

$$\Omega_{\text{stack}}(q) = |q|^2 = e^{-2s}$$

from the APG cusp-regularized metric. The main theorem proves

$$\left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_2^2 \leq C_1 H(W)^2.$$

Consequently,

$$\rho_{\text{APG}} \in L^2(X), \quad \|\rho_{\text{APG}}\|_2 < \infty,$$

including cusp neighborhoods. The result removes the source-blow-up obstruction and reduces Open Problem 15.1 to the remaining spectral-gap and conductor-growth estimate

$$\lambda_1(\Delta_{\text{APG}})^{-1} H(W)^2 \leq C \deg(\pi) \log N.$$

The paper does not claim an independent proof of Fermat’s Last Theorem, the *abc* conjecture, the Szpiro conjecture, or the full APG–Arakelov Projection Theorem.

Keywords: Arithmetic Power Geometry; source concentration; Shannon entropy; Rényi entropy; modular curves; Arakelov geometry; Hilbert–Schmidt operators; cusp integrability; spectral gap; Dirichlet energy.

1 Introduction

Arithmetic Power Geometry (APG) studies algebraic closure relations under exponent deformation. The arithmetic setting of APG lies within the broader framework of classical algebraic number theory and Diophantine arithmetic (Borevich and Shafarevich, 1966). Classical arithmetic geometry usually fixes polynomial exponents and studies rational or integral points over arithmetic fields; APG reverses this perspective by fixing positive base coordinates and treating the exponent as a deformation parameter. The first APG volume established that the local closure-defect rate near the Euclidean target is governed by Shannon entropy (Akhtar, 2026a; Shannon, 1948; Cover and Thomas, 2006). APG II developed an information-geometric stabilization of the corresponding parameter space (Akhtar, 2026b; Amari, 2016). APG III introduced the integrated closure-defect functional A_0 , while APG IV and APG V introduced scale correction, coordinate damping, and discrete prime-chain regularization (Akhtar, 2026c,d,e).

APG VI and APG VII connected this analytic deformation architecture to Arakelov geometry by formulating a conditional APG–Arakelov height-coupling program and a discrepancy-source compatibility framework on compactified modular curves (Akhtar, 2026f,g; Arakelov, 1974; Faltings, 1984).

The surrounding Diophantine and Arakelov-height background follows the classical literature on arithmetic geometry, height theory, and Diophantine approximation (Faltings, 1983; Lang, 1983; Vojta, 1987; Soulé, 1992; Zhang, 1993; Bombieri and Gubler, 2006; Hindry and Silverman, 2000; Silverman, 1986, 2007).

The present paper should be viewed as part of the broader Arithmetic Power Geometry research program. APG I established entropy-governed local closure-defect behavior near the Euclidean target (Akhtar, 2026a); APG II developed the information-geometric stabilization framework (Akhtar, 2026b); APG III introduced the integrated closure-defect functional A_0 (Akhtar, 2026c); APG IV analyzed coordinate-dependent defect structures and modularity obstructions (Akhtar, 2026d); APG V developed discrete moduli-stack regularization and boundary-control mechanisms (Akhtar, 2026e); APG VI formulated a conditional APG–Arakelov height-coupling program (Akhtar, 2026f); APG VII established discrepancy-source compatibility within the APG–Arakelov architecture (Akhtar, 2026g); and APG VIII derived the Spectral–Green Bridge relating APG energy, source discrepancy, and spectral control (Akhtar, 2026h). The objective of APG IX is to prove the source-concentration component required for the APG spectral-conductor program. APG VIII then proved the Spectral–Green Bridge

$$E_{\text{APG}} \leq \lambda_1(\Delta_{\text{APG}})^{-1} \left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_2^2. \quad (1)$$

This bridge reduces the analytic side of the APG projection program to two independent tasks: control of the source fluctuation and control of the spectral gap. The present paper proves the source-control half.

The main theorem of APG IX is

$$\left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_2^2 \leq C_1 H(W)^2. \quad (2)$$

Here $H(W)$ is the Shannon entropy of the APG Euclidean weight vector, and C_1 is a structural constant depending only on the APG source-normalization, transport-kernel, and deformation-window constants.

The proof has four components. First, we construct the source decomposition without circularly assuming the desired estimate. Second, we prove a Variance–Defect Transfer Theorem using a Hilbert–Schmidt transport kernel. Third, we use Shannon–Rényi control to bound the integrated defect by entropy. Fourth, we prove cusp L^2 -control by defining the APG cusp-regularized metric with canonical quadratic vanishing at the boundary divisor.

2 Preliminaries and APG Setting

2.1 The modular surface

Let

$$X = X_0(N)^*(\mathbb{C})$$

denote the compactified modular curve of level N . Modular curves, compactifications, and their arithmetic moduli interpretations are treated classically by [Hecke \(1928\)](#); [Shimura \(1971\)](#); [Deligne \(1971\)](#); [Katz \(1973\)](#); [Mazur \(1977, 1978\)](#); [Katz and Mazur \(1985\)](#); [Diamond and Shurman \(2005\)](#). The treatment of compactification and boundary-divisor phenomena is closely related to classical studies of compact models of arithmetic quotients and noncompact modular spaces ([Mumford, 1977](#)). We view X as a compact Riemann surface equipped with a finite APG-compatible measure $d\mu$. The arithmetic-surface and intersection-theoretic framework underlying later APG–Arakelov constructions draws on classical work in Arakelov geometry and arithmetic surfaces ([Arakelov, 1974](#); [Szpiro, 1981](#); [Saito, 1988](#); [Faltings, 1984](#); [Abbes and Ullmo, 1997](#)). The geometric language used throughout APG is compatible with the modern algebraic-geometric framework of schemes, divisors, and morphisms developed in classical algebraic geometry ([Hartshorne, 1977](#)). Write

$$V = \text{Vol}(X) = \int_X d\mu < \infty.$$

The Hilbert space $L^2(X, d\mu)$ is equipped with

$$\langle f, g \rangle = \int_X f(z) \overline{g(z)} d\mu(z), \quad \|f\|_2^2 = \int_X |f(z)|^2 d\mu(z).$$

The zero-mean subspace is

$$L_0^2(X) = \left\{ f \in L^2(X, d\mu) : \int_X f d\mu = 0 \right\}.$$

2.2 APG weights and entropy

For fixed positive base coordinates $a, b > 0$, define

$$w_a = \frac{a^2}{a^2 + b^2}, \quad w_b = \frac{b^2}{a^2 + b^2}. \quad (3)$$

Then $W = (w_a, w_b)$ is a probability vector. Its Shannon entropy is

$$H(W) = -w_a \log w_a - w_b \log w_b. \quad (4)$$

The APG local closure defect at the Euclidean target is

$$\delta_K(t) = \left| 1 - \frac{a^t + b^t}{(a^2 + b^2)^{t/2}} \right|. \quad (5)$$

APG I proves the local entropy-governed asymptotic

$$\delta_K(t) = \frac{H(W)}{2} |t - 2| + O((t - 2)^2) \quad (t \rightarrow 2), \quad (6)$$

under its Euclidean-target normalization (Akhtar, 2026a). The Shannon–Rényi control paper strengthens this into a fixed-window bound of the form

$$A_0(a, b; n) \leq C_\delta H(W)^2, \quad (7)$$

where A_0 is the integrated closure-defect functional and $C_\delta > 0$ depends only on the chosen deformation window (Akhtar, 2026i; Rényi, 1961).

3 APG IX Axioms and Definitions

The local algebraic structures underlying APG deformation spaces are consistent with standard constructions in commutative algebra (Atiyah and Macdonald, 1969). To make the proof self-contained inside APG IX, we explicitly state the analytic source axioms used in the theorem.

APG Axiom 3.1 (APG probability source). *The APG deformation source is a probability density*

$$\rho_{\text{APG}} = \frac{d\mathbb{P}_{\text{APG}}}{d\mu}$$

with respect to $d\mu$, satisfying

$$\rho_{\text{APG}} \geq 0, \quad \int_X \rho_{\text{APG}} d\mu = 1.$$

APG Axiom 3.2 (APG local mismatch field). *There exists a measurable local mismatch field*

$$\mathcal{D}_{\text{local}}(\cdot; n) \in L^2(X, d\mu)$$

such that

$$A_0(a, b; n) = \int_X |\mathcal{D}_{\text{local}}(z; n)|^2 d\mu(z).$$

APG Axiom 3.3 (Transport-source realization). *The fluctuation source field is generated by a transport response to the local mismatch:*

$$F_{\text{APG}}(z) = \int_X K_{\text{trans}}(z, \zeta) \mathcal{D}_{\text{local}}(\zeta; n) d\mu(\zeta).$$

The kernel K_{trans} is the zero-mean generalized Green transport kernel of the elliptic APG Laplacian on X .

APG Axiom 3.4 (Cusp-regularized APG metric). *Near a cusp $\mathfrak{c}$, let*

$$q = e^{2\pi iz}, \quad s = -\log |q|.$$

The APG cusp-regularized metric is defined by the conformal rule

$$g_{\text{APG}} = |q|^2 g_{\text{hyp}}.$$

Equivalently, the APG cusp multiplier is

$$\Omega_{\text{stack}}(q) = |q|^2 = e^{-2s}.$$

Remark 3.5. The word “unconditional” in this paper means unconditional after these APG IX definitions and APG I–VIII source conventions are adopted. It does not mean independent of the APG framework or independent of all prior APG definitions.

4 Non-Circular Intrinsic Source Construction

Definition 4.1 (Fluctuation source). *Define*

$$F_{\text{APG}} = \rho_{\text{APG}} - \frac{1}{V}. \tag{8}$$

Lemma 4.2 (Mean-zero property). *The fluctuation source satisfies*

$$\int_X F_{\text{APG}} d\mu = 0.$$

Proof. Using the normalization of ρ_{APG} ,

$$\int_X F_{\text{APG}} d\mu = \int_X \left(\rho_{\text{APG}} - \frac{1}{V} \right) d\mu = 1 - \frac{1}{V} \int_X d\mu = 1 - \frac{1}{V} V = 0.$$

□

Lemma 4.3 (Fluctuation-space admissibility). *Under the APG transport-source realization axiom, $F_{\text{APG}} \in L_0^2(X)$.*

Proof. The mean-zero property was proved above. For square-integrability, use

$$F_{\text{APG}}(z) = \int_X K_{\text{trans}}(z, \zeta) \mathcal{D}_{\text{local}}(\zeta; n) d\mu(\zeta).$$

The kernel K_{trans} is Hilbert–Schmidt by [theorem 5.1](#) below, and $\mathcal{D}_{\text{local}} \in L^2(X)$ by the definition of A_0 . Hence the integral operator maps $L^2(X)$ to $L^2(X)$, so $F_{\text{APG}} \in L^2(X)$. Since it also has mean zero, $F_{\text{APG}} \in L_0^2(X)$. $\square$

Definition 4.4 (Amplitude and discrepancy direction). *Define*

$$\alpha_{\text{APG}} = \|F_{\text{APG}}\|_2. \quad (9)$$

If $\alpha_{\text{APG}} > 0$, *define*

$$\sigma = \frac{F_{\text{APG}}}{\|F_{\text{APG}}\|_2}. \quad (10)$$

If $\alpha_{\text{APG}} = 0$, *set* $\sigma = 0$.

Proposition 4.5 (Intrinsic source decomposition). *The APG source admits the decomposition*

$$\rho_{\text{APG}} = \frac{1}{V} + \alpha_{\text{APG}} \sigma, \quad (11)$$

with

$$\int_X \sigma d\mu = 0$$

and

$$\|\sigma\|_2 = 1$$

whenever $\alpha_{\text{APG}} > 0$.

Proof. By definition,

$$\rho_{\text{APG}} = \frac{1}{V} + F_{\text{APG}}.$$

If $F_{\text{APG}} \neq 0$, then

$$F_{\text{APG}} = \|F_{\text{APG}}\|_2 \frac{F_{\text{APG}}}{\|F_{\text{APG}}\|_2} = \alpha_{\text{APG}} \sigma.$$

Thus (11) follows. Since $F_{\text{APG}} \in L_0^2(X)$, $\int_X \sigma d\mu = 0$. Also,

$$\|\sigma\|_2 = \left\| \frac{F_{\text{APG}}}{\|F_{\text{APG}}\|_2} \right\|_2 = 1.$$

If $F_{\text{APG}} = 0$, the decomposition holds with $\alpha_{\text{APG}} = 0$ and $\sigma = 0$. $\square$

The analytic framework employed below is consistent with classical complex analysis and compact Riemann surface theory ([Ahlfors, 1979](#)).

5 Kernel Construction and Variance-Defect Transfer

Theorem 5.1 (Kernel existence and Hilbert–Schmidt finiteness). *Let Δ_{APG} be the elliptic Laplace–Beltrami operator on the compact APG surface X , restricted to $L_0^2(X)$. Let*

$$0 < \lambda_1 \leq \lambda_2 \leq \dots$$

be its positive eigenvalues and let $\{\phi_n\}_{n \geq 1}$ be an orthonormal eigenbasis of $L_0^2(X)$. Define

$$K_{\text{trans}}(z, \zeta) = \sum_{n=1}^{\infty} \lambda_n^{-1} \phi_n(z) \overline{\phi_n(\zeta)}. \quad (12)$$

Then $K_{\text{trans}} \in L^2(X \times X, d\mu \otimes d\mu)$, and

$$M_0 = \int_X \int_X |K_{\text{trans}}(z, \zeta)|^2 d\mu(z) d\mu(\zeta) = \sum_{n=1}^{\infty} \lambda_n^{-2} < \infty.$$

Proof. The APG surface is compact after cusp regularization, and Δ_{APG} is elliptic and self-adjoint on $L_0^2(X)$. Therefore it has a discrete positive spectrum. By the spectral theorem, the inverse Green operator on $L_0^2(X)$ is given by

$$Gf = \sum_{n=1}^{\infty} \lambda_n^{-1} \langle f, \phi_n \rangle \phi_n.$$

Its kernel is (12). The Hilbert–Schmidt norm is

$$\|G\|_{HS}^2 = \sum_{n=1}^{\infty} \lambda_n^{-2}.$$

By Weyl’s law for compact two-dimensional Riemannian manifolds (Buser, 1992; Iwaniec, 2002),

$$\lambda_n \sim C_X n \quad (n \rightarrow \infty)$$

for some $C_X > 0$. Hence

$$\sum_{n=1}^{\infty} \lambda_n^{-2} \ll \sum_{n=1}^{\infty} n^{-2} < \infty.$$

Therefore $K_{\text{trans}} \in L^2(X \times X)$, and $M_0 < \infty$. □

Theorem 5.2 (Variance–Defect Transfer). *Let F_{APG} be generated by the transport-source realization*

$$F_{\text{APG}}(z) = \int_X K_{\text{trans}}(z, \zeta) \mathcal{D}_{\text{local}}(\zeta; n) d\mu(\zeta).$$

Then

$$\alpha_{\text{APG}}^2 = \|F_{\text{APG}}\|_2^2 \leq M_0 A_0(a, b; n), \quad (13)$$

where

$$M_0 = \|K_{\text{trans}}\|_{HS}^2 < \infty.$$

Proof. For each $z \in X$, the Cauchy–Schwarz inequality gives

$$|F_{\text{APG}}(z)|^2 \leq \left(\int_X |K_{\text{trans}}(z, \zeta)|^2 d\mu(\zeta) \right) \left(\int_X |\mathcal{D}_{\text{local}}(\zeta; n)|^2 d\mu(\zeta) \right).$$

By definition,

$$\int_X |\mathcal{D}_{\text{local}}(\zeta; n)|^2 d\mu(\zeta) = A_0(a, b; n).$$

Thus

$$|F_{\text{APG}}(z)|^2 \leq \left(\int_X |K_{\text{trans}}(z, \zeta)|^2 d\mu(\zeta) \right) A_0(a, b; n).$$

Integrating over z yields

$$\|F_{\text{APG}}\|_2^2 \leq \left(\int_X \int_X |K_{\text{trans}}(z, \zeta)|^2 d\mu(\zeta) d\mu(z) \right) A_0(a, b; n).$$

The double integral is M_0 , finite by [theorem 5.1](#). Therefore

$$\alpha_{\text{APG}}^2 \leq M_0 A_0(a, b; n).$$

□

Theorem 5.3 (Entropy-amplitude bridge). *Assume the Shannon–Rényi APG control estimate*

$$A_0(a, b; n) \leq C_\delta H(W)^2.$$

Then

$$\alpha_{\text{APG}} \leq C_H H(W),$$

where

$$C_H = \sqrt{M_0 C_\delta}.$$

Proof. By [theorem 5.2](#),

$$\alpha_{\text{APG}}^2 \leq M_0 A_0(a, b; n).$$

Using the Shannon–Rényi bound gives

$$\alpha_{\text{APG}}^2 \leq M_0 C_\delta H(W)^2.$$

Taking square roots gives

$$\alpha_{\text{APG}} \leq \sqrt{M_0 C_\delta} H(W).$$

Set $C_H = \sqrt{M_0 C_\delta}$.

□

6 APG Cusp Regularization and L^2 Control

Theorem 6.1 (APG cusp regularization). *Let $q = e^{2\pi iz}$ be the local cusp coordinate on $X_0(N)^*(\mathbb{C})$, and set*

$$s = -\log |q|.$$

Define the APG cusp-regularized metric by

$$g_{\text{APG}} = |q|^2 g_{\text{hyp}}.$$

Then the stack multiplier is

$$\Omega_{\text{stack}}(q) = |q|^2 = e^{-2s}.$$

Proof. Near a cusp, the cusp divisor is $q = 0$. APG cusp regularization assigns to this divisor the canonical quadratic vanishing weight

$$\Omega_{\text{stack}}(q) = |q|^2.$$

Since

$$s = -\log |q|,$$

we have

$$|q| = e^{-s}.$$

Squaring gives

$$|q|^2 = e^{-2s}.$$

Hence

$$\Omega_{\text{stack}}(q) = |q|^2 = e^{-2s}.$$

□

Definition 6.2 (APG cusp measure). *In a cusp cylinder with coordinates (s, θ) , the APG regularized cusp measure is*

$$d\mu_{\text{APG}} = e^{-2s} ds d\theta. \tag{14}$$

Theorem 6.3 (Cusp L^2 -integrability). *Let $U_{\mathfrak{c}}$ be a cusp neighborhood. Suppose that for some finite $\beta \geq 0$,*

$$|\sigma(s, \theta)| \leq C_{\mathfrak{c}} s^{\beta}$$

on $U_{\mathfrak{c}}$. Then

$$\sigma \in L^2(U_{\mathfrak{c}}, d\mu_{\text{APG}}).$$

Proof. Using (14),

$$\|\sigma\|_{L^2(U_{\mathfrak{c}})}^2 = \int_0^{2\pi} \int_{s_0}^{\infty} |\sigma(s, \theta)|^2 e^{-2s} ds d\theta.$$

The growth condition gives

$$\|\sigma\|_{L^2(U_{\mathfrak{c}})}^2 \leq 2\pi C_{\mathfrak{c}}^2 \int_{s_0}^{\infty} s^{2\beta} e^{-2s} ds.$$

Set $x = 2s$. Then $ds = dx/2$, and

$$\int_{s_0}^{\infty} s^{2\beta} e^{-2s} ds = \frac{1}{2^{2\beta+1}} \int_{2s_0}^{\infty} x^{2\beta} e^{-x} dx.$$

The special-function estimates used below are classical and may be found in standard references on analysis and special functions (Whittaker and Watson, 1927). The remaining integral is the upper incomplete gamma function

$$\Gamma(2\beta + 1, 2s_0),$$

which is finite for finite $\beta \geq 0$ and $s_0 > 0$. Hence

$$\|\sigma\|_{L^2(U_\epsilon)}^2 < \infty.$$

□

Corollary 6.4 (Global L^2 -admissibility). *If σ is smooth on the compact interior of X and satisfies polynomial-logarithmic growth at each cusp, then*

$$\sigma \in L^2(X, d\mu_{\text{APG}}).$$

Consequently,

$$F_{\text{APG}} \in L^2(X), \quad \rho_{\text{APG}} \in L^2(X).$$

Proof. There are finitely many cusps. The interior contribution is finite because σ is smooth on compact subsets. Each cusp contribution is finite by theorem 6.3. Therefore $\sigma \in L^2(X)$. Since

$$F_{\text{APG}} = \alpha_{\text{APG}} \sigma$$

and $\alpha_{\text{APG}} < \infty$ by theorem 5.3, we obtain $F_{\text{APG}} \in L^2(X)$. Finally,

$$\rho_{\text{APG}} = \frac{1}{V} + F_{\text{APG}},$$

and $1/V \in L^2(X)$ because $V < \infty$. Hence $\rho_{\text{APG}} \in L^2(X)$. □

7 Main Source Concentration Theorem

Theorem 7.1 (APG Source Concentration Bound). *Under the APG IX source, transport, entropy-control, and cusp-regularized metric definitions above,*

$$\rho_{\text{APG}} \in L^2(X), \quad \|\rho_{\text{APG}}\|_2 < \infty,$$

including cusp neighborhoods, and

$$\left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_2^2 \leq C_1 H(W)^2, \tag{15}$$

where

$$C_1 = C_H^2 = M_0 C_\delta.$$

Proof. By the intrinsic decomposition,

$$\rho_{\text{APG}} - \frac{1}{V} = F_{\text{APG}} = \alpha_{\text{APG}} \sigma.$$

If $F_{\text{APG}} = 0$, then (15) is immediate. Otherwise, $\|\sigma\|_2 = 1$. Hence

$$\left\| \rho_{\text{APG}} - \frac{1}{V} \right\|_2^2 = \|\alpha_{\text{APG}} \sigma\|_2^2 = \alpha_{\text{APG}}^2 \|\sigma\|_2^2 = \alpha_{\text{APG}}^2.$$

By theorem 5.3,

$$\alpha_{\text{APG}}^2 \leq C_H^2 H(W)^2.$$

Therefore

$$\left\| \rho_{\text{APG}} - \frac{1}{V} \right\|_2^2 \leq C_H^2 H(W)^2.$$

Set $C_1 = C_H^2 = M_0 C_\delta$. The global L^2 -membership of ρ_{APG} , including cusp neighborhoods, follows from theorem 6.4. $\square$

Corollary 7.2 (No APG source blow-up). *The APG source fluctuation cannot blow up independently of the entropy profile:*

$$\left\| \rho_{\text{APG}} - \frac{1}{V} \right\|_2 \leq \sqrt{C_1} H(W).$$

Proof. Take the square root of (15). $\square$

8 Consequence for the Spectral–Conductor Program

Combining theorem 7.1 with the APG VIII Spectral–Green Bridge (1) gives the following result.

Corollary 8.1 (Spectral–entropy energy bound).

$$E_{\text{APG}} \leq C_1 \lambda_1(\Delta_{\text{APG}})^{-1} H(W)^2.$$

Proof. By APG VIII,

$$E_{\text{APG}} \leq \lambda_1(\Delta_{\text{APG}})^{-1} \left\| \rho_{\text{APG}} - \frac{1}{V} \right\|_2^2.$$

By theorem 7.1,

$$\left\| \rho_{\text{APG}} - \frac{1}{V} \right\|_2^2 \leq C_1 H(W)^2.$$

Combining gives the result. $\square$

Corollary 8.2 (Reduction of Open Problem 15.1). *Open Problem 15.1 is reduced to proving*

$$\lambda_1(\Delta_{\text{APG}})^{-1} H(W)^2 \leq C' \deg(\pi) \log N.$$

Proof. The target inequality is

$$\lambda_1(\Delta_{\text{APG}})^{-1} \left\| \rho_{\text{APG}} - \frac{1}{V} \right\|_2^2 \leq C \deg(\pi) \log N.$$

By theorem 7.1, it is enough to prove

$$C_1 \lambda_1(\Delta_{\text{APG}})^{-1} H(W)^2 \leq C \deg(\pi) \log N.$$

Equivalently,

$$\lambda_1(\Delta_{\text{APG}})^{-1} H(W)^2 \leq C' \deg(\pi) \log N$$

with $C' = C/C_1$. □

9 Discussion

APG IX proves the source-side theorem required by the APG spectral-conductor program. The proof has been made internally complete by stating the APG IX source, transport, entropy-control, and cusp-regularized metric definitions explicitly before deriving the main estimate. The theorem does not prove Open Problem 15.1 completely because the spectral factor

$$\lambda_1(\Delta_{\text{APG}})^{-1}$$

still requires independent conductor-scale control. That problem belongs to APG X–XII. The representation-theoretic foundations of the modularity program also draw on the seminal work of [Serre \(1987\)](#). The broader modularity and Fermat-context literature includes the Frey-curve program, modularity lifting theory, and the proof of Fermat’s Last Theorem ([Frey, 1986](#); [Ribet, 1990](#); [Diamond, 1996](#); [Taylor and Wiles, 1995](#); [Wiles, 1995](#); [Darmon et al., 1997](#)). The contribution of APG IX is therefore precise: it eliminates the possibility that the source term

$$\rho_{\text{APG}} - \frac{1}{V}$$

is the obstruction. The remaining APG program must now focus on spectral gap stability and conductor domination.

10 Conclusion

This paper proves the APG Source Concentration Bound:

$$\left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_2^2 \leq C_1 H(W)^2.$$

It also proves

$$\rho_{\text{APG}} \in L^2(X), \quad \|\rho_{\text{APG}}\|_2 < \infty,$$

including cusp neighborhoods under the APG cusp-regularized metric. The proof proceeds through a non-circular source decomposition, a Hilbert–Schmidt kernel construction, a Variance–Defect Transfer Theorem, a Shannon–Rényi entropy-amplitude bridge, and an explicit APG cusp regularization theorem proving

$$\Omega_{\text{stack}}(q) = |q|^2 = e^{-2s}.$$

Consequently, APG IX achieves its stated source-side objective and reduces Open Problem 15.1 to the remaining spectral-conductor inequality

$$\lambda_1(\Delta_{\text{APG}})^{-1}H(W)^2 \leq C \deg(\pi) \log N.$$

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